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abstracts

[2025/2018](#) (PDF)

Last updated: 2026-10-02

Batched and Packed (Publicly) Verifiable Secret Sharing: A Unified Framework and Applications

Shahla Atapoor, Karim Baghery, Georgio Nicolas, Robi Pedersen, and Jannik Spiessens

[Hide abstract](#) -

Cryptographic protocols

Verifiable Secret Sharing (VSS) allows a dealer to distribute a secret among n parties so that each can verify their share's validity, and any qualified subset can reconstruct the secret. Publicly Verifiable Secret Sharing (PVSS) extends VSS by enabling anyone to verify the correctness of distributed shares. Both VSS and PVSS schemes are core building blocks in many cryptographic applications. We introduce a k -batched and l -packed extension of Π , a unified framework from PKC 2025 for Shamir-based computational VSS in the synchronous setting with optimal resilience. Our framework enables the sharing and verification of $l \times k$ secrets in a single protocol execution, offering a tunable trade-off

[2026/1443](#) (PDF)

Last updated: 2026-10-02

SENTRA: Privacy-Preserving Training in Outsourced Cloud Environments

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Cryptographic protocols

Training machine learning models in untrusted clouds requires strong guarantees of confidentiality, integrity, and correctness, while remaining scalable and resilient to node churn. These challenges are further amplified in emerging agentic AI systems, where autonomous and distributed learning components require trustworthy coordination and secure state management across heterogeneous cloud environments. Existing Trusted Execution Environments (TEEs) lack scalability and remain vulnerable to side-channel attacks for large workloads, while pure secure multi-party computation (MPC) approaches incur prohibitive overhead in practice. SENTRA (Secure ENclave-based TRaining

[2026/075 \(PDF\)](#)

Last updated: 2026-10-02

From TS-SUF-2 to TS-SUF-4: Practical Security Enhancements for FROST2 Threshold Signatures

Will Wang, Syh-Yuan Tan, Ryan Chow, Chanson Chan, and Martin Zhao

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Public-key cryptography

Threshold signature schemes play a vital role in securing digital assets within blockchain and distributed systems. FROST2 stands out as a practical threshold Schnorr signature scheme, noted for its efficiency and compatibility with standard verification processes. However, under the one-more discrete logarithm assumption, with static corruption and centralized key generation settings, FROST2 has been shown by Bellare et al. (in CRYPTO 2022) to achieve only

2026/2141 (PDF)

Last updated: 2026-10-01

BitZ: proofs and commitments in arbitrary rings through binary fields

Remco Bloemen, Albert Garreta, Marcin Kostrzewa, Shreyas Londhe, and John Wu

[Hide abstract](#) -

Cryptographic protocols

We introduce BitZ, a hash-based Polynomial Commitment Scheme (PCS) for committing to multilinear polynomials f with coefficients in an arbitrary finitely generated ring S , e.g. a finite field $\mathbb{F}$, the integers $\mathbb{Z}$, a cyclotomic ring, etc. Moreover, given another arbitrary ring R and a ring homomorphism $\psi : S \rightarrow R$, BitZ then proves evaluation claims over R for the polynomial $\psi(f)$. BitZ's costs depend almost exclusively on the number of bits in the coefficients of f , and not on S , R or ψ . Moreover, BitZ provides range checks (or more generally, bit-size checks) essentially for free.

BitZ can thus be used as a PCS in essentially any proof system. We do so to build a SNARK, called BitZ-SNARK, for integer polynomial constraints, following the fingerprinting technique of Campanelli and Hall-Andersen, where one commits over $\mathbb{Z}$ and proves the constraints over a random prime field $\mathbb{F}_q$, i.e. BitZ is deployed with $S = \mathbb{Z}$, $R = \mathbb{F}_q$, and

2026/1997 (PDF)

Last updated: 2026-10-01

Cryptology ePrint Archive

Hosein Hadipour, Saleh Khalaj Monfared, Jens Aich, and Jan Vorloeper

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Secret-key cryptography

The Walsh-Hadamard transform (WHT) computes the correlation of a Boolean function with every linear function at once. Cryptanalytic workflows transform large arrays repeatedly, in batches. However, available tools specialize in particular data types, platforms, or ecosystems. We present LibFWHT, an open-source C library for dense WHTs, with Python bindings and a command-line interface. It includes vectorized, multicore, and graphics processing unit (GPU) backends, batch operations, and routines for Boolean functions and substitution boxes (S-boxes). With one central processing unit (CPU) core, LibFWHT is 2.5 to 3.3 times faster than FFTW; on batched Boolean spectra with eight threads, it is up to 3.7 times faster than sboxU. On an NVIDIA H200, its device-resident GPU backend reaches 1.8 trillion operations per second and runs the same Boolean workload about 600 times faster than the CPU-only sboxU. We demonstrate LibFWHT in linear and differential-linear cryptanalysis of

[2026/122 \(PDF\)](#)

Last updated: 2026-10-01

The Motte-and-Bailey Framework for Leakage-Resilient Accordion Modes: Featuring Qaitbay and Alicante

Mario Marhuenda Beltrán and Mustafa Khairallah

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Secret-key cryptography

Accordion modes have experienced a surge in popularity, partially motivated by the recent NIST Accordion modes

leakage-resilient Accordion mode. We start by presenting a generic analysis of the Encode-then-Encipher (EtE) framework in the leakage-resilient setting, assuming the enciphering is a leakage-resilient STPRP (STPRPI2). We show that the resulting security, while strong, suffers from some limitations. Next, we introduce Motte-and-Bailey, a general framework for building

[2025/286](#) (PDF)

Last updated: 2026-10-01

Verifiable Computation for Approximate Homomorphic Encryption Schemes

Ignacio Cascudo, Anamaria Costache, Daniele Cozzo, Dario Fiore, Antonio Guimarães, and Eduardo Soria-Vazquez

[Hide abstract](#) -

Cryptographic protocols

We address the problem of proving the validity of computation on ciphertexts of homomorphic encryption (HE) schemes, a feature that enables outsourcing of data and computation while ensuring both data privacy and integrity. We propose a new solution that handles computations in RingLWE-based schemes, particularly the CKKS scheme for approximate arithmetic. Our approach efficiently handles ciphertext arithmetic in the polynomial ring R_q without emulation overhead and manages ciphertexts maintenance operations, such as modulus switching, key switching, and

[2026/2145](#) (PDF)

Last updated: 2026-10-01

Fully Anonymous Perfect Secret-Sharing

Roni Con, Ray Li, and Noam Mazor

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Foundations

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or every unauthorized set of participants are uniformly random, and every authorized set can reconstruct the secret exactly using only the share values, without needing to know which participant holds which share or in what order the shares appear. It was previously open whether such schemes exist for nontrivial exact-threshold parameters $2 < t < n$. For the smallest previously open threshold, $t = 3$, we give, for every $n \geq 4$, an explicit fully anonymous scheme for one-bit

[2026/1795 \(PDF\)](#)

Last updated: 2026-10-01

On Removing Interaction from Quantum Proofs

Nicholas Spooner and Max Tromanhauser

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Foundations

An important open question in quantum cryptography is the construction of publicly-verifiable NIZKs for QMA. Classically, one can construct NIZKs for NP in the random oracle model (and sometimes in the standard model) by compiling an honest-verifier ZK (HVZK) Σ -protocol for NP using the Fiat-Shamir transformation. Broadbent and Grilo introduced a

[2026/1457 \(PDF\)](#)

Last updated: 2026-10-01

IBE and PIR from Isogenies

Shweta Agrawal, Andrea Basso, and Sikhar Patranabis

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Public-key cryptography

We significantly expand the reach of isogeny-based cryptography by constructing, for the first time, two fundamental primitives that have long remained out of reach from isogenies: identity-based encryption (IBE) and semi-

additionally achieves anonymity, which means that the ciphertexts hide both the message and the recipient identity.

Our results are enabled by new techniques for constructing cryptographic primitives from structured isogeny computations, centered around the first construction of blind batch encryption from isogenies. To establish its security, we introduce a new hardness assumption, CDH with Mismatched Torsion (CDHwMT), which informally captures the hardness of

[2026/2256 \(PDF\)](#)

Last updated: 2026-10-01

Affine-Padding Rabin-Oracle Factoring in $L_n\left(\frac{1}{3}, \sqrt[3]{\frac{32}{9}}\right)$

Alexandra-Ioana Buzătoiu, Diana Maimuț, and David Naccache

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Attacks and cryptanalysis

The 2007 algorithm of Joux, Naccache, and Thomé (\textsc{jnt}) shows that, with subexponential access to an e -th root oracle, one can forge \textsc{rsa} signatures for \emph{odd} e in time close to the special number field sieve, without factoring the modulus; a recent \textsc{jnt} implementation by Shea et al.~\cite{Forge26} carried that attack to 1024-bit \textsc{rsa}. We revisit the \textsc{jnt} construction at $e = 2$, the Rabin case, where an e -th root oracle is a square-root oracle. A \emph{raw} square-root oracle factors $n = pq$ in one query, so it is of

oracle that returns a single canonical root and thereby resists the one-query attack. We show

that a one-sided number field sieve against such an oracle produces a congruence of squares

$X^2 \equiv Y^2 \pmod{n}$ with $X \not\equiv \pm Y$ with probability $\frac{1}{2}$ per dependency, and

hence \emph{factors} n in special-number-field-sieve time

$$L_n\left(\frac{1}{3}, \sqrt[3]{\frac{32}{9}}\right) \simeq L_n(1/3, 1.526 \dots),$$

[2026/2232](#) (PDF)

Last updated: 2026-10-01

Cryptanalysis of the ICCS NGCC Round-1 Public-Key Candidates

Zhenyu Xiong and Mingsheng Wang

[Hide abstract](#) -

Public-key cryptography

The ICCS Next-Generation Commercial Cryptography (NGCC) round-1 call received 84 public-key candidates for public evaluation. We present a design-level assessment that targets weaknesses no local code fix can close, identifying eight such flaws. Each recurs a pitfall already known from NIST PQC

[2026/777](#) (PDF)

Last updated: 2026-10-01

How Strong is the FO-Calypse, Really? Instantiating Plaintext-Checking Oracles against Masked Software Implementations of ML-KEM

Brieuc Balon, Gaëtan Cassiers, Thibaud Schoenauen, and François-Xavier Standaert

(PCOs) instantiated thanks to the leakage of the re-encryption step taking place during decapsulation are a well-known weakness of ML-KEM. An already wide literature investigated how to efficiently exploit such oracles, leading to easy (full) key recoveries. Somewhat surprisingly, the investigation of how to best instantiate PCOs against ML-KEM's most leaking operations is less investigated, in particular when it comes to quantitative evaluations against concrete masked implementations. In this paper, we first remedy this lack by systematically instantiating PCOs against three open source masked software implementations of the Keccak function used in ML-KEM, based on different masking

2024/1383 (PDF)

Last updated: 2026-10-01

Self-Orthogonal Minimal Codes From (Vectorial) p -ary Plateaued Functions

René Rodríguez Aldama, Enes Pasalic, Fengrong Zhang, and Yongzhuang Wei

[Hide abstract -](#)

Foundations

In this article, we derive the weight distribution of linear codes stemming from a subclass of (vectorial) p -ary plateaued functions (for a prime p), which includes all the explicitly known examples of weakly and non-weakly regular plateaued functions. This construction of linear codes is referred in the literature as the first generic construction. First, we partition the class of p -ary plateaued functions into three classes C_1 , C_2 , and C_3 , according to the behavior of their dual function f^* . Using these classes, we refine the results

derive the full weight distributions of codes stemming from

[2024/1583](#) (PDF)

Last updated: 2026-10-01

Efficient Pairing-Free Adaptable k-out-of-N Oblivious Transfer Protocols

Keykhosro Khosravani, Taraneh Eghlidos, and Mohammad reza Aref

[Hide abstract](#) -

Cryptographic protocols

Oblivious Transfer (OT) is one of the fundamental building blocks in cryptography that enables various privacy-preserving applications. Constructing efficient OT schemes has been an active research area. This paper presents three efficient two-round pairing-free k-out-of-n oblivious transfer protocols with standard security. Our constructions follow the minimal communication pattern: the receiver sends k

[2026/2271](#) (PDF)

Last updated: 2026-10-01

Exact SVP on Haar-Random Lattices at the Heuristic Sieving Exponents

Yang Zhou, Zhedong Wang, and Mingsheng Wang

[Hide abstract](#) -

Foundations

We give a Monte Carlo algorithm for exact Euclidean SVP on Haar-random lattices with arithmetic time $(3/2)^{n/2+o(n)}$ and space $(4/3)^{n/2+o(n)}$, up to polynomial factors in the logarithmic condition number of the input basis. The leading exponents match those of heuristic lattice sieving. For a set of lattices of Haar measure $1 - o(1)$, the algorithm succeeds with probability $1 - o(1)$ on every input basis.

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Markov chain on the nonzero lattice points in a smaller ball.

[2026/431](#) [\(PDF\)](#)

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Revisiting the Security of Sparkle

Ojaswi Acharya, Georg Fuchsbauer, Adam O'Neill, and Marek Sefranek

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Public-key cryptography

We revisit the three-round threshold Schnorr signature scheme Sparkle of Crites, Komlo, and Maller (CRYPTO 2023), as well as its variant Sparkle+. While Sparkle+ was accompanied by a claim of full adaptive security, subsequent work identified a gap in the analysis. The original—and simpler and more efficient—Sparkle scheme has so far lacked even a proof of static security.

We resolve this by proving static security for Sparkle; our main result is then a tight proof of its full adaptive security in the pure random oracle model, i.e., without relying on the algebraic group model. The core obstacle is that, in the fully adaptive setting, rewinding arguments fundamentally break

[2026/2205](#) [\(PDF\)](#)

Last updated: 2026-09-30

DKG Is All You Need

Guru-Vamsi Policharla

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Cryptographic protocols

We construct the first Batched Threshold Encryption scheme with a *transparent* setup where public parameters are *independent* of the batch size. As a result batches of

assumption -- the decisional bilinear square Diffie--Hellman assumption.

Setup is just a distributed key generation protocol to sample secret shares of a random value. Ciphertexts consist of two $\mathbb{G}_T$ elements: one $\mathbb{G}_T$ element and the encrypted message

[2026/1547 \(PDF\)](#)

Last updated: 2026-09-30

Silent Distributed Cryptography for DNFs and Threshold Policies from Lattices

Abtin Afshar, Rishab Goyal, and Saikumar Yadugiri

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Foundations

We study two central problems in threshold cryptography from lattices: (1) threshold encryption with silent setup for general thresholds $t \geq 2$, where no post-quantum constructions were previously known, and (2) threshold fully homomorphic encryption (TFHE) with sublinear parameters, an open problem since the work of [Boneh et al.; CRYPTO'18].

We introduce

(α, β) -Scaled Linear Secret Sharing Schemes (LSSS), a relaxation of standard LSSS in which each authorized set S reconstructs a scaled version of the secret, $\gamma_S \cdot k$, where both the reconstruction coefficients and the set-dependent scaling factor γ_S are bounded over the integers. Unlike prior approaches based on bit decomposition, this preserves the uniform distribution of unauthorized shares. Building on this,



Encryption from Lattices

Rishab Goyal and Saikumar Yadugiri

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Public-key cryptography

Broadcast encryption (BE) allows a sender to succinctly encrypt a message to any dynamically chosen subset of recipients. The gold-standard for BE is *optimal succinctness* (parameters independent of the number of users) and *adaptive security*. Attaining both from falsifiable post-quantum assumptions has been a central open problem. Recently, Goyal and Yadugiri (GY) [Goyal-Yadugiri; CRYPTO'26] gave the first adaptively-secure and optimally-succinct slotted distributed BE under a falsifiable lattice assumption, but their techniques inherently require an a-priori bound on the number of users and a slotted user structure. Two highly-sought-after generalizations thus

[2026/792 \(PDF\)](#)

Last updated: 2026-09-30

Equivocal Broadcast Encryption: Adaptively-Secure Optimal Distributed Broadcast Encryption from Lattices

Rishab Goyal and Saikumar Yadugiri

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Public-key cryptography

We present the first Distributed Broadcast Encryption (DBE) scheme from falsifiable lattice assumptions that achieves adaptive security with optimal parameters (short public/secret keys and ciphertexts). Our construction enjoys transparent setup and offers flexible instantiation: we achieve a succinct CRS in the Random Oracle Model, or a long CRS in

Refining Probabilistic-Linearization TIDA for 5-Round SHA3-384 Collision Attacks (Full Version)

Luhan Yan, Zhenzhen Bao, and Huina Li

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Attacks and cryptanalysis

Since the SHA-3 family was standardized by NIST in 2015, its collision resistance has been extensively studied. The previous best-known collision attack on 5-round SHA3-384 is based on internal differentials and the probabilistic-linearization variant of two-block Target Internal Differential Algorithm (TIDA). Its connector replaces deterministic affine restrictions that guarantee S-box differential validity by higher-dimensional affine relaxations, reducing the number of linear constraints and preserving more degrees of freedom. The price is that solutions of the linearized system are only probabilistically valid. In particular, once the first message

GlueLUT: Efficient Lookup Table Arguments over Residue Rings

Yuanju Wei, Zhelei Zhou, Xinxuan Zhang, Songyu Wu, Binwu Xiang, Cheng Hong, Yi Deng, and Xiaofeng Chen

[Hide abstract](#) -

Cryptographic protocols

Lookup table polynomial interactive oracle proofs (LUT PIOPs) make SNARK arithmetizations concise, yet almost all efficient constructions assume field arithmetic. We show that directly instantiating them over a composite residue ring $\mathbb{Z}_Q$ can be unsound: component-wise set checks preserve the residue

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call the CKI alignment ambiguity.

To overcome this ambiguity, we present GlueLUT, a family of efficient LUT PIOPs over residue rings. GlueLUT runs the lookup over an auxiliary field while retaining the arithmetic

[2026/1862 \(PDF\)](#)

Last updated: 2026-09-30

HEAT: Faster Fully Homomorphic Inference via Approximations-Weights Co-Adaptation

Alessandro Zirilli, Davide Marincione, Evgenios M.

Kornaropoulos, Giuseppe Ateniese, and Emanuele Rodolà

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Applications

Fully homomorphic encryption (FHE) allows a server to run a language model directly on encrypted user prompts, but current approaches remain prohibitively slow. Ciphertexts natively support only addition, multiplication, and rotation, and multiplications may be composed only to a bounded depth before a costly bootstrapping operation is required to continue. Every nonlinearity must therefore be approximated by an iterative method; each iteration increasing the number of multiplications. A higher iteration count buys precision but

[2024/1906 \(PDF\)](#)

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On Efficient Computations of $y^2 = x^3 + b/\mathbb{F}_p$ for Primes $p \equiv 1 \pmod{3}$

Guangwu Xu, Wei Yu, Ke Han, and Pengfei Lu

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Public-key cryptography

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multiplication on KOBLITZ curves over binary fields. A long-standing open problem has been to identify a suitable family of elliptic curves over prime fields for which the window τ -NAF approach can be effectively extended. In this paper, we settle this problem by establishing such an extension for the family $E_b : y^2 = x^3 + b$ over $\mathbb{F}_p$ with prime $p \equiv 1 \pmod{3}$. This family includes several practically important curves used in blockchain applications (e.g., secp256k1) and pairing-based cryptography (e.g., BN254 and BLS12-381). By

[2026/2237](#) (PDF)

Last updated: 2026-09-30

Lower Bounds on Random-Oracle-Model Signature Length

Alon Chayet, Iftach Haitner, Mathias Marty, and Eylon Yogev

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Foundations

Hash-based signatures offer a conservative foundation for post-quantum authentication, but their large signatures impose substantial communication and storage costs. For security parameter λ , discrete-logarithm-based signatures have length $O(\lambda)$, while known hash-based constructions have quadratic signature length, up to logarithmic factors, even for one-time signing. Is this gap inherent?

We prove what are, to the best of our knowledge, the first lower bounds on the length of signature schemes in the (pure) random oracle model. Our bounds apply to schemes with non-adaptive verifiers and a natural security property called salted soundness: a scheme with such security is

[2024/1626](#) (PDF)

Last updated: 2026-09-30

Faster Proofs and VRFs from Isogenies

Shai Levin and Robi Pedersen

[Hide abstract](#) -

Cryptographic protocols

We improve recent generic proof systems for isogeny knowledge by Cong, Lai, Levin based on circuit satisfiability, by using radical isogeny descriptions to prove a path in the underlying isogeny graph. We then present a new generic construction for a verifiable random function (VRF) based on a one-more type hardness assumption and zero-knowledge

[2026/2274](#) (PDF)

Last updated: 2026-09-30

zk-BAN: An efficient anonymous blocklisting system with signature-based revocation

Kosei Akama, Yoshimichi Nakatsuka, Koichi Moriyama, and Keisuke Uehara

[Hide abstract](#) -

Applications

Anonymous blocklisting enables services to revoke malicious users while preserving anonymity for honest users. Signature-based revocation supports revocation from past authentications, but existing systems incur substantial overhead at large scale and for users returning after long offline periods. Window-based alternatives reduce this overhead but impose restrictive revocation-timing policies.

[2026/2273](#) (PDF)

Last updated: 2026-09-30

Yinsuan Wu

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Cryptographic protocols

Witness encryption over blockchain state aims to release a secret on a ledger condition without a key custodian.

Validators with enough stake to construct a valid fork witness can obtain the secret although the condition never holds on the canonical chain. They finalize a private fork that satisfies the condition, and keeping the fork hidden withholds the conflicting votes that slashing needs. We propose slashable secrecy to address this gap with Ethereum's native slashing rules, without an additional key-custody committee or separate collateral. In executions where the condition never

2026/2275 (PDF)

Last updated: 2026-09-30

Defeating Time-Average Selfish Mining Across Epoch Boundaries in Nakamoto Consensus

Ren Zhang

[Hide abstract](#) -

Cryptographic protocols

Selfish mining profits in Nakamoto consensus because difficulty adjustment mechanisms (DAMs) misinterpret uncounted orphans as hash-rate contraction. Although orphan-aware DAMs record orphaned proof-of-work, strict same-epoch inclusion rules enable an orphan exclusion attack (OEA): an adversary can race private forks near epoch boundaries to permanently exclude honest tail orphans from retargeting, restoring substantial profits in short-epoch protocols. We propose the pipelined buffer difficulty

Pseudorandom Functions

Yaobin Shen, Lei Wang, and Dawu Gu

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Secret-key cryptography

Key derivation functions can be used to derive variable-length random strings that serve as cryptographic keys. They are integral to many widely-used communication protocols such as TLS, IPsec and Signal. NIST SP 800-108 specifies several key derivation functions based on pseudorandom functions such as CMAC and HMAC, that can be used to derive additional keys from an existing cryptographic key. This standard either explicitly or implicitly requests their KDFs to be variable output length pseudorandom function, collision resistant, and preimage resistant, which are also demanded by practical applications. Yet, since the publication of this standard dating back to the year of 2008, until now, there is no formal analysis to justify these security properties of KDFs.

[2025/395 \(PDF\)](#)

Last updated: 2026-09-30

On the (In)security of Approximate Computation Protocols from CKKS

Intak Hwang, Yisol Hwang, Miran Kim, Dongwon Lee, and Yongsoo Song

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Public-key cryptography

Secure Multi-Party Computation (MPC) enables collaborative and privacy-preserving computation over private inputs. Recent advances in Homomorphic Encryption (HE),

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world applications involving approximate arithmetic. However, the inherent errors introduced by CKKS pose substantial challenges in the design and analysis of CKKS-based secure computation protocols.

In this work, we investigate the correctness and security of existing CKKS-based MPC protocols. Most prior constructions rely on the noise smudging technique, in which each party independently adds exponentially large noise to the output to guarantee the security of the protocol. However, we show that this approach does not achieve standard simulation-

[2026/2272](#) (PDF)

Last updated: 2026-09-30

Attacking UOV-based Signatures over divided power algebras

Peigen Li and Siyong Tao

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Public-key cryptography

Merz and Ran derive additional equations from divided powers of the public polar forms in characteristic two.

[2026/2270](#) (PDF)

Last updated: 2026-09-29

UC PAKE and Concrete Security

Tingfei Feng and Jiayu Xu

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Foundations

The Universal Composability (UC) framework provides a paradigm for security definitions that allows for arbitrary composition, but has been criticized for being unamicable to

2026/2269 (PDF)

Last updated: 2026-09-29

Consensus in One Shot: Fast Agreement Beyond Classical Limits

Omri Shmueli and Ilan Tennenhouse

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Cryptographic protocols

Good-case latency, studied in early fast-consensus protocols (Martin and Alvisi, DSN 2005) and formalized by Abraham, Nayak, Ren, and Xiang (DISC 2020), captures a natural efficiency goal: when the sender is honest, agreement should be reached quickly for the honest parties. In the standard classical model, however, there exists a barrier for any $f \geq n/3$ corrupted parties. Specifically, synchronous broadcast requires good-case latency at least $\Delta + \delta$ for $f \geq n/3$, where δ is the actual, unknown message-delay bound and $\Delta \gg \delta$ is its known, conservative upper bound. Under partial synchrony, consistency and liveness cannot even both hold for $f \geq n/3$ (Dwork, Lynch, and Stockmeyer, JACM 1988).

In this work we study the use of quantum information to get

2026/2268 (PDF)

Last updated: 2026-09-29

Kopis: A KEM for Obfuscation

Andrea Basso and Michael Rosenberg

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Public-key cryptography

Password-authenticated key exchange (PAKE) and obfuscated key exchange (OKEX) are widely used protocols, appearing in

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increasing number of proposals for post-quantum PAKE and OKEX. All such protocols are similar in that they build on a KEM, obfuscating public keys and/or ciphertexts sent over the wire, e.g., by adding a random mask or applying an ideal cipher. Many propose instantiation with ML-KEM, a NIST-standardized lattice-based KEM.

Unfortunately, ML-KEM is a poor fit for these obfuscating protocols. The algebraic structure of ML-KEM public keys and ciphertexts means that obfuscating these values requires custom procedures for hash-to-vector, matrix expansion, serialization/deserialization, and randomized ciphertext decompression. In order to be broadly useful, these procedures also must be constant-time to prevent side channels and must be standardized to permit interoperability.

[2026/2267 \(PDF\)](#)

Last updated: 2026-09-29

Impersonation Resilience of Subversion-Resilient UC Protocols

Paula Arnold, Sebastian Berndt, Jörn Müller-Quade, Astrid Ottenhues, and Johannes Ottenhues

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Cryptographic protocols

Subversion attacks where the attacker aims to replace parts of a cryptographic system by manipulated parts in an undetectable manner have been shown to model both theoretical and practical attacks such as supply chain attacks. Originally proposed by Young and Yung (CRYPTO 1996), interest in them has increased dramatically after the

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and Rogaway (CRYPTO 2014).

One of the most promising countermeasures against such attacks are cryptographic reverse firewalls, proposed by Mironov and Stephens-Davidowitz (EUROCRYPT 2015), which are small devices used by the parties with the aim to rerandomize the protocol message to remove possible leakage. A wide range of protocols and corresponding

[2026/2266](#) (PDF)

Last updated: 2026-09-29

Chinese NGCC Algorithms: The First Week of AI Cryptanalysis

Markku-Juhani O. Saarinen

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Attacks and cryptanalysis

On September 20, 2026, the Chinese Institute of Commercial Cryptography Standards published the 119 first-round candidates of its Next-generation Commercial Cryptographic Algorithms (NGCC) program: 34 signature schemes, 41 key encapsulation mechanisms, 9 key exchange protocols, and 35 hash functions. The program follows the template of the open NIST competitions for AES, SHA-3, and post-quantum cryptography. It opened just after cryptographers' "Lee Sedol moment" in the summer of 2026, when AI analysis overtook humans in analyzing the HAWK signature scheme. During the

[2026/2062](#) (PDF)

Last updated: 2026-09-29

Hamming Ideals and Grobner Bases for ISD-like Syndrome Decoding

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We investigate an algebraic approach to the Syndrome Decoding Problem, based on a reformulation of the Hamming weight constraint and its integration with the Information Set Decoding paradigm. We begin with a systematic analysis of the Hamming variety, deriving its defining equations in terms of elementary symmetric functions. Since these equations may have high degree, we exploit convolution identities for elementary symmetric functions, together with factorizations based on Lucas' identity, to derive an equivalent formulation with auxiliary variables and equations of bounded degree.

Building on this modeling, we generalize the ISD paradigm through an ISD-like decoding strategy, implemented by the

[2025/1606 \(PDF\)](#)

Last updated: 2026-09-29

Collatz Hash: Hash Algorithm Using $3x+1$ Conjecture

Shaurya Pratap Singh, Bhupendra Singh, and Alok Mishra

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Cryptographic protocols

In this paper, we introduce a new hash algorithm in that we used the Collatz problem, focusing on its conditional branching structure, an element often overlooked despite the fame of the $3x + 1$ conjecture. This hash algorithm takes an arbitrary-length input and produces a fixed length 512/384/256 bit output. The presented hash algorithm is in

[2026/2265 \(PDF\)](#)

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Simona Ceskova, Martin Peresini, and Ivan Homollak

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Applications

In public blockchains, the ability to order visible transactions allows block producers to extract value from users, a problem known as Maximal Extractable Value (MEV). Proposer-Builder Separation has turned that right into a formalized auction, and the leading protocol-level mitigations answer it with payload blindness: encrypted mempools, verifiable permutations, and block delay hide or scramble transactions before the order is fixed. Every such mitigation carries a security argument that assumes an adversary who learns nothing before the order commits. Deployed systems face a different adversary, and this paper measures the cost of that

[2026/2264 \(PDF\)](#)

Last updated: 2026-09-29

THEMIS: A Co-Designed System for Encrypted Transformer Inference

Shihao Li, Wenhao Wang, Lu Li, Xiaomei Tang, Jian Liu, Rongmao Chen, Cheng Hong, and Guangfu Sun

[Hide abstract -](#)

Applications

Secure Transformer inference can be made non-interactive under fully homomorphic encryption (FHE), but the computation is extremely expensive. Existing FHE-based systems reduce that cost by optimizing individual stages in isolation, each within its own component rather than against the CKKS level budget they all share. A local gain therefore may not translate into an end-to-end speedup.

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principles, one bounding the depth each kind of stage may spend and one fixing where on the modulus chain it runs. THEMIS generalizes coefficient-encoded plaintext-ciphertext matrix multiplication (PCMM) to slot-encoded ciphertexts

[2026/2263](#) (PDF)

Last updated: 2026-09-29

Levis: Extension-Free Multi-Key Fully Homomorphic Encryption in the Plain Model

Baoyu Li, Binwu Xiang, and Kang Yang

[Hide abstract](#) -

Public-key cryptography

Multi-key fully homomorphic encryption (MKFHE) enables computation over ciphertexts encrypted under keys independently generated by different parties. Recently, Min, Park, and Song (MPS, CRYPTO 2026) constructed the first RLWE-based MKFHE scheme in the plain model, i.e., without trusted or interactive setup. Their construction requires each local compact Gentry-Sahai-Waters (GSW) ciphertext to be extended to a joint-key GSW-like ciphertext before evaluation, incurring $O(dk^2)$ gadget products per input for k parties and gadget length d .

[2026/2262](#) (PDF)

Last updated: 2026-09-29

NOMOS: Secure Non-Interactive k NN under CKKS

Shihao Li, Wenhao Wang, Xiaomei Tang, Jian Liu, Rongmao Chen, Lu Li, zhiqiang chen, and Guangfu Sun

[Hide abstract](#) -

Applications

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under fully homomorphic encryption remains expensive because it requires distance computation, encrypted ranking, and top- k extraction. CKKS is attractive for this setting because it natively supports packed approximate arithmetic, yet existing CKKS-based systems such as Engorgio (USENIX Security 2025) incur high overhead when used for reranking candidate sets.

We present NOMOS, an encrypted k NN protocol for candidate sets that targets the ranking and top- k extraction

[2026/2261](#) (PDF)

Last updated: 2026-09-29

A Provable Subexponential-Time Algorithm for LWE from $n + o(n)$ Samples via Wagner-Style Gaussian Sampling

Xuyuan Han, Yiming Gao, and Honggang Hu

[Hide abstract](#) -

Attacks and cryptanalysis

We give a randomized subexponential-time algorithm for decision Learning with Errors (LWE) in two complementary parameter regimes. For polynomial prime modulus $q = n^{\kappa+o(1)}$ with any fixed $\kappa > 1/2$, the algorithm handles independent discrete Gaussian errors drawn from $D_{\mathbb{Z}, s_e}$, with Gaussian parameter s_e satisfying $\ln(2 + s_e) = (\ln n)^{a+o(1)}$ for any fixed $0 \leq a < 1$. Conversely, polynomial error $s_e = n^{\gamma+o(1)}$, for any fixed $\gamma > 0$, can be handled with a quasipolynomial prime modulus $q = \exp((\ln n)^{1+\eta+o(1)})$ for any fixed $\eta > 0$. In both regimes, for an arbitrary secret distribution, the algorithm succeeds with probability $1 - o(1)$ using $m = n + \Theta(n/\ln \ln n)$ original samples, with

2026/197 (PDF)

Last updated: 2026-09-29

Derivative-Based Evaluation of Multivariate Polynomials over Product Hamming Balls

Vaibhav Dixit, Santanu Sarkar, Fukang Liu, and Willi Meier

[Hide abstract -](#)

Foundations

Efficient evaluation of multivariate polynomials over finite spaces is an important problem in algebraic cryptanalysis, particularly in exhaustive-search attacks on multivariate public-key cryptosystems. Existing fast exhaustive-search techniques, including those of Bouillaguet et al. (CHES 2010), Dinur (EUROCRYPT 2021), and Furue and Takagi (PQCrypto 2023), mainly target the complete space $\mathbb{F}_q^n$, whereas many cryptanalytic applications require evaluation only over



2026/1163 (PDF)

Last updated: 2026-09-29

SWIFT: Shallow and SIMD-Aware CKKS Functional Bootstrapping for Low-Latency

Jung Hee Cheon, Daehyun Jang, and Jaehee Kang

[Hide abstract -](#)

Public-key cryptography

The CKKS homomorphic encryption scheme can achieve high throughput for large batches, typically of tens of thousands of inputs, whereas DM/CGGI schemes offer low latency for a single input.

In existing CKKS approaches, the multiplicative depth required for nonlinear function evaluation demands a large ciphertext modulus, which forces a large ring degree

require

only a few hundred function evaluations at a time.

For these moderate batch sizes, existing CKKS-based methods

requiring large multiplicative depth incur high latency without fully benefiting from their high throughput.

We present **SWIFT**, a CKKS functional-bootstrapping method that reduces latency by using multiple slots to lower multiplicative depth.

For input x , **SWIFT** packs its integer multiples ℓx into unused slots. Exponential bootstrapping then generates the powers $\exp(2\pi i \ell x) = \exp(2\pi i x)^\ell = \alpha^\ell$ in parallel.

Compared to previous approaches that compute these powers through homomorphic multiplications, **SWIFT** directly generates those powers and reduces the required multiplicative depth and ciphertext modulus, allowing a smaller ring degree and lower latency.

We also introduce a trigonometric approximation method for singular functions such as $1/\sqrt{x}$, whose derivative is unbounded near zero.

We observe that the previous Fourier extension method of Bian et al.

(EUROCRYPT'26) can produce extremely large coefficient

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Hannes Hacker, Shekoufen Nelsarian, Sara Zain, and Elif Bulge Kavun

[Hide abstract](#) -

Public-key cryptography

Migrating blockchain systems to Post-Quantum

Cryptography (PQC) requires not only replacing classical primitives, but also analysing transformations that change how signatures are represented and verified. We study Falcon-PKR, a public-key-recovery variant of Falcon, motivated by Zcash-like transparent transaction workflows. Falcon-PKR replaces explicit storage of the full Falcon public key with a compact stored digest and reconstructs a candidate public key from the signature during verification. In our construction, this reduces the combined representation size of public key and signature by approximately 15%, but it also introduces a distinct security case in which verification depends on a reconstructed key derived from adversarially

[2026/1958](#) (PDF)

Last updated: 2026-09-29

Arithmetic for Large-Characteristic Finite Fields in CKKS

Daehyun Jang and Junho Lee

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Public-key cryptography

Seur'e and Suvanto (ePrint 2026/1102) recently showed that, for small-characteristic primes p , arithmetic over $\mathbb{F}_{p^r}$ can be homomorphically evaluated in CKKS via a technique they call *spectral encoding*.

Their construction is, however, restricted to small characteristic: ciphertext multiplication amplifies the error by

encoded to a plaintext with operator norm at most $\mathcal{O}(rp)$.

The error

therefore grows by a factor of rp in the worst case, limiting the supported size of the characteristic prime p .

In this work, we propose *bi-spectral encoding*, which decomposes the

coefficients of the plaintext polynomials used in the spectral encoding of Seur'e and Suvanto.

With a decomposition parameter d , a field element in $\mathbb{F}_{p^r}$ can be encoded to a plaintext with operator norm $\mathcal{O}(rdp^{1/d})$, for a suitable choice of encoding parameters.

[2026/2259](#) (PDF)

Last updated: 2026-09-29

Supersingularity and Superspeciality Verification of Abelian Surfaces

Maria Corte-Real Santos, Gioella Lorenzon, and Krijn Reijnders

[Hide abstract](#) -

Public-key cryptography

Supersingular abelian surfaces are essential in isogeny-based cryptography. Despite this, we have no efficient algorithm to verify if a given abelian surface is supersingular. In this work, we initiate this research topic by giving an efficient Monte Carlo algorithm to verify if an abelian surface over $\mathbb{F}_p$ is

[2026/2258](#) (PDF)

Last updated: 2026-09-29

Superposition Key-Recovery Attacks on Dilithium and Fiat-Shamir Signature Schemes

Carlos Cid, David Elkouss, and Manuel Goulão

scheme standardised by NIST, ML-DSA, and its precursor identification schemes under an extended quantum adversarial model, one in which the adversary is allowed to use quantum resources to interact with a prover/signer. We begin our analysis by developing new techniques for superposition attacks that realise a relative phase oracle for the LSBs of a classical function evaluated in a quantum computer. We then extend these techniques to enable a relative phase oracle for arbitrary bits. Leveraging these

[2026/2257](#) (PDF)

Last updated: 2026-09-29

Screaming Channel: Recent Advances in Far Field EM Side-Channel Attacks based on Radio Coupling

Huanyu Wang, Pei Cao, Chi Zhang, and Dawu Gu

[Hide abstract](#) -

Attacks and cryptanalysis

The Internet of Things (IoT) is revolutionizing society by enabling anytime, anywhere connectivity, but its widespread deployment also makes IoT devices attractive targets for malicious activities. Over the past decade, radio-coupling based Far-field EM Side-Channel Attacks (FEM-SCAs), also known as screaming channel attacks, have emerged as a new threat to these wireless IoT edges. Unlike conventional power or near-field EM side-channel attacks, FEM-SCAs exploit the unintended coupling between digital computation and the Radio Frequency (RF) transmission chain, allowing sensitive information to be captured through radiated radio signals at long distances. Since the first screaming-channel demonstration, this field has progressed rapidly: attacks have

Signature Scheme for One-Time Issuance Identities

Zhiwei Wang

[Hide abstract](#) -

Cryptographic protocols

We present an identity-based blind signature scheme built on the SM9 Chinese national cryptographic standard. The scheme

outputs standard SM9 signatures, removes all online pairing operations through a single setup-time pre-computation, and adds only one user-side exponentiation to achieve blindness.

We prove two properties rigorously: $\text{\emph{restricted target-identity unforgeability}}$ (RTU) under the Q -BCAA1 assumption, and $\text{\emph{computational blindness}}$ against a malicious signer in the random oracle model (ROM), where the

signer is given the final signatures and still cannot link them to its protocol views. We state clearly that RTU is strictly weaker than the standard one-more unforgeability (OMU) notion for blind signatures: RTU forbids signing queries

on the target identity and is therefore meaningful only when that identity encodes a fresh, one-time issuance context, as occurs in coin-, ballot-, and credential-issuance protocols.

The Q -BCAA1 assumption is the inversion-type assumption that

matches the SM9 private-key structure

$d_{\text{\textcolor{red}{ID}}} = [s/(s + \tau)]P_1$; it lets the reduction answer

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Yuanxin Wang, Chengcheng Chang, and Kai Hu

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Attacks and cryptanalysis

We improve differential-linear distinguishers and derive key-recovery attacks on reduced-round versions of the PRFs Orthros, Gleeok, ZIP-AES, ZIP-GIFT-64, and ZIP-GIFT-128. Relative to previous differential-linear analyses, we extend the reported coverage for Orthros from 7 to 9 rounds and for Gleeok-128 from 4 to 5 rounds. For Orthros and Gleeok-128, we obtain estimated correlation magnitudes $2^{-61.40}$ and $2^{-31.71}$, respectively. We also obtain distinguishers for 6-round Gleeok-256, (3, 3)-round ZIP-AES, (7, 7)-round ZIP-

[2026/1240 \(PDF\)](#)

Last updated: 2026-09-29

Adaptive attacks on FESTA variants with masked-degree isogenies

Tomoki Moriya

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Attacks and cryptanalysis

FESTA is an isogeny-based trapdoor function proposed as a high-performance trapdoor function in isogeny-based cryptography. Its core design principles have inspired a number of related constructions, collectively referred to as FESTA variants.

The MOXZ attack is an adaptive attack that exploits malicious ciphertexts together with access to a checking oracle, aiming to compromise FESTA and its variants. This attack applies to

[2026/978 \(PDF\)](#)

Last updated: 2026-09-29

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Shune Cao, Xin Wang, Chenxu Wang, Xiao Sui, and Sisi Duan

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Cryptographic protocols

TLA⁺ (Temporal Logic of Actions) is a formal specification language well-suited for distributed systems. However, writing proper TLA⁺ scripts requires high domain expertise. When it comes to modeling Byzantine behaviors for Byzantine fault-tolerant consensus protocols, the simulation of malicious behavior is a fundamental challenge: overly simplified modeling misses critical vulnerabilities, and verbose modeling leads to state-space explosion.

In this paper, we present TLAssist, a large language model

[2026/2152 \(PDF\)](#)

Last updated: 2026-09-29

Structural Analysis of Seven Hash Functions Submitted to the NGCC

Yufei Yuan, Ruichen Wu, Shanpeng Wei, Junxu Shen, Jinpeng Liu, and Yixin Zhang

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Secret-key cryptography

The Institute of Commercial Cryptography Standards (ICCS) launched the Next-generation Commercial Cryptographic Algorithms Program (NGCC) and invited worldwide comments on draft submission requirements and evaluation criteria for cryptographic hash algorithms. Our analysis of seven submitted hash functions gives the following results:

- Message differences that cancel in every key injection give explicit collisions for all four fixed-output variants of

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- Two distinct states of Neulaser become equal after one update, giving collisions for all three variants with the initialization used by the v2 reference and optimized implementations.
- An invariant subspace of CHIME-512 permits collision search using at most 2^{64} hash evaluations when shifts act separately on 64-bit words, as in both submitted implementations.
- CHAMP 's determinant constraint gives collision

[2026/2219](#) (PDF)

Last updated: 2026-09-29

Incrementally Verifiable Computation without Extraction

Abhishek Jain, Surya Mathialagan, and Brent Waters

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Cryptographic protocols

Incrementally verifiable computation (IVC) [Valiant, TCC '08] allows one to iteratively prove that a configuration x_0 reaches a configuration x_T via T repeated applications of a (possibly non-deterministic) machine M . An IVC scheme is fully succinct if the proof size is independent of both T and the size of the intermediate configurations.

In this work, we develop a new indistinguishability obfuscation (iO)-based approach to IVC that avoids the extraction-based security analyses central to prior constructions. Assuming subexponential hardness of iO and

[2026/1740](#) (PDF)

Last updated: 2026-09-28

Prabhanjan Ananth

[Hide abstract -](#)

Foundations

Non-local search-to-decision asks whether the difficulty of two non-communicating (non-local) parties both predicting x given a bipartite state (that possibly depends on x) implies the difficulty of non-local parties both predicting the inner product $\langle r, x \rangle$, for a uniformly random r . This problem and its variants have been extensively studied and are motivated

[2026/2254 \(PDF\)](#)

Last updated: 2026-09-28

AEBAP: An Efficient ECC-Based Authentication Protocol Designed for Wireless Networks

Reza Hooshmand and Sajjad Alizadeh

[Hide abstract -](#)

Cryptographic protocols

The Internet of Things (IoT) comprises a vast array of interconnected devices that rely heavily on wireless communication to enable applications in domains such as healthcare, transportation, and smart environments. However, intrinsic vulnerabilities in wireless networks, such as susceptibility to eavesdropping, impersonation, and message tampering, require the development of robust yet lightweight authentication mechanisms. To address these concerns, we introduce AEBAP, an efficient authentication protocol based on elliptic curve cryptography

[2026/563](#) (PDF)

Last updated: 2026-09-28

Optimizing FROST for Message Capacity

Philipp Jovanovic, Ben Riva, and Arnab Roy

[Hide abstract](#) -

Cryptographic protocols

The FROST threshold signature scheme achieves round optimal Schnorr signing through a double-nonce construction, but requires two presignatures per signature. Since each presignature demands an expensive distributed key generation (DKG) protocol, this overhead is significant for high-throughput applications. FROST builds on a core presignature protocol (that we call FROST-core) that uses hash-based re-randomization of presignatures. We investigate whether fewer presignatures can be used to sign multiple messages, improving FROST-core's message capacity.

[2026/2215](#) (PDF)

Last updated: 2026-09-28

Quantum security analysis of unrestricted isogeny-based group actions

Xavier Bonnetain, Max Duparc, Dania Lazzarini, Luciano Maino, Chloe Martindale, Christophe Petit, Sina Schaeffler, André Schrottenloher, and Alessandro Sferlazza

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Attacks and cryptanalysis

Setting concrete parameters for group action based cryptographic protocols is notoriously difficult due to the incomplete understanding of the concrete efficiency of Kuperberg's subexponential quantum attack. Estimates for

recent and far more efficient alternative group actions.

[2026/454](#) (PDF)

Last updated: 2026-09-28

The principal ideal problem for endomorphism rings of superspecial abelian varieties

Wouter Castryck, Jonathan Komada Eriksen, Riccardo Invernizzi, and Frederik Vercauteren

[Hide abstract](#) -

Foundations

We describe a Las Vegas algorithm for the principal ideal problem in matrix rings $M_g(O)$ for $g \geq 2$, over maximal orders O in the rational quaternion algebra $B_{p,\infty}$ ramified at ∞ and a prime number p . Under plausible heuristic

[2025/722](#) (PDF)

Last updated: 2026-09-28

One-Step Schnorr Threshold Identification

Foteinos Mergoupis-Anagnostou

[Hide abstract](#) -

Cryptographic protocols

Threshold cryptographic primitives have not been widely adopted in real-world distributed systems (i.e., beyond the closed committee model) presumably due to state-synchronization overhead and complex certification processes for the shareholders. These are both aspects of infrastructure overreliance, an assumption that is usually glossed over in their design. In this work, we propose OSST, a real-time threshold identification protocol that achieves non-interactivity and non-reliance on public shares by means of direct proof interpolation. Given a Shamir (n, t) -shared secret x , the proposed scheme allows any $t^* \geq t$ (but no less)

Case of Inverse Diffie-Hellman

Akshima, Eike Kiltz, Aysan Nishaburi, Samin Nooripoor, and Emiel Wiedijk

[Hide abstract -](#)

Foundations

In a multi-instance problem, the adversary is given n independent problem instances and succeeds if it solves at least k of them.

We study the generic hardness of a broad class of multi-instance problems over prime-order groups, including multi-instance Computational Diffie-Hellman (CDH), Square Diffie-Hellman (SDH), Inverse Diffie-Hellman (IDH), and Linear Kernel Diffie-Hellman (LKDH). As expected, we show that, in generic groups, solving multi-instance CDH and SDH is as hard as solving k independent discrete logarithm instances. In contrast, although the single-instance versions of CDH and IDH are known to be equivalent, we establish generic upper

[2026/2252 \(PDF\)](#)

Last updated: 2026-09-28

Batch Decryption from New Standard Assumptions

Nico Döttling, Bernardo Magri, Benjamin Marsh, and Mahesh Sreekumar Rajasree

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Public-key cryptography

Batch decryption allows an authority to release a short key that enables public decryption of a selected batch of ciphertexts. Motivated by encrypted

authorization semantics and assumptions.

First, we provide a simple and compact epochless construction in an RSA group based on Fiat's Batch RSA [CRYPTO'89]. Its batch-decryption hint is just one group element, and its keys and ciphertexts have size independent of the batch size, with no batch bound fixed at setup. We prove adaptive security under the strong RSA assumption in the random-oracle model. The scheme permits repeated releases for overlapping batches, with each release authorizing decryption of the selected ciphertexts.

[2026/2251](#) (PDF)

Last updated: 2026-09-28

Adaptively UC-Secure Oblivious Transfer from Group Actions

Olivier Blazy, Céline Chevalier, Saqib Kakvi, and Lola-Baie Mallordy

[Hide abstract](#) -

Cryptographic protocols

Oblivious Transfer (OT) is a fundamental building block for secure computation, in which a sender holds two messages and a receiver learns

exactly one of them, without revealing which one was chosen and

without learning the other. While generic constructions for Universally Composable (UC) OT exist, instantiating them from post-quantum assumptions remains challenging, particularly under adaptive corruption.

weakly pseudorandom Effective Group Actions (EGA) like CSIDH. We adhere to the framework of [BC15], which derives an adaptively UC-secure OT protocol from SPHF-friendly extractable and equivocable (E^2) commitment schemes.

[2026/2250 \(PDF\)](#)

Last updated: 2026-09-28

Decryption Failures in NGCC Lattice KEMs: Correlated Blocks, Omitted Compression Noise, and Failure Boosting under a Query Cap

Yuyang Xiao

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Public-key cryptography

The Chinese NGCC post-quantum competition received a family of lattice KEMs whose decryption failure rates (DFRs) are certified by their designers with models that differ in detail. We recompute the failure probability of the first-round lattice KEMs whose claimed DFR, or a simple recomputation of it, lies near or below the nominal level, and we estimate the offline search for weak ciphertexts under a cap of 2^{64} or 2^{80} decapsulation queries. Three distinct mechanisms are relevant.

(i)~\emph{Correlated decoding blocks.} DTRU decodes 16-coefficient DoubleE₈ blocks; in the tricyclotomic ring

are ring-paired and the repetition code places the same bit on both. Our

covariance-aware estimates raise the DFR of six of the seven sets by roughly

50--110 bits. Their estimated uncapped first-failure costs are 2^{85} -- 2^{111} decapsulations. Under a fitted ciphertext-tail model, DTRU-648, DTRU-768, and DTRU-Prime fit within 2^{64}

expected queries after

2^{22} -- 2^{26} estimated offline encapsulations.

(ii)~\emph{Omitted compression noise.} Cheetah compresses its public key and

both ciphertext components. Section~2.2 of the specification omits the

public-key term $\langle e_b, r \rangle$, while the v -compression error can be treated as a known, ciphertext-dependent shift. Under the coefficient-independence model, exponentially tilted one-dimensional FFTs give

DFR estimates of $2^{-79.1}$ and $2^{-51.8}$ for Cheetah128 and Cheetah256,

rather than the claimed 2^{-129} and 2^{-176} . Their expected random-query waiting times are below 2^{80} . We check the noise model at

assumption in post-quantum cryptography. Regev established a quantum reduction from LWE to the Dihedral Coset Problem (DCP). Later, Brakerski et al. introduced the Extrapolated Dihedral Coset Problem (EDCP), proving its equivalence to LWE. However, unlike DCP, EDCP no longer admits a coset structure. This limits the direct application of techniques for hidden subgroup problems.

In this work, we introduce the Cyclotomic Coset Problem (CCP), a cyclotomic generalization of DCP that preserves an exact hidden-subgroup structure. Let ζ_p be a primitive p -th root of unity, let $\pi = \zeta_p - 1$, and write $q = p^t$ and $L = t(p - 1)$.

[2026/2248](#) (PDF)

Last updated: 2026-09-28

ML-DSA masking sweetened with SUCRE: Shuffle-and-Unmask Countermeasure for REjection sampling

Sonia Belaïd, Ryad Benadjila, Julien Devevey, Morgane Guereau, Thomas Legavre, Ange Martinelli, Thomas Ricosset, Matthieu Rivain, and Mélissa Rossi

[Hide abstract](#) -

[Implementation](#)

We present SUCRE, a novel countermeasure designed to physically protect the rejection sampling step of ML-DSA, one of the post-quantum signature schemes standardized by NIST. At the core of SUCRE is a masking gadget that securely unmaskes a vector while simultaneously applying a random permutation of its coefficients. This lightweight mechanism preserves the vector's infinity norm, enabling rejection

We formally prove that a d -probing adversary can learn at most some permuted rejected values---information which, we show, should remain insufficient to endanger the security

[2026/1884](#) (PDF)

Last updated: 2026-09-28

Correction of parameters in ``Computing Optimal Ate Pairings on Elliptic Curves with Embedding Degree 9, 15 and 27''

Walid Haddaji

[Hide abstract](#) -

Foundations

We correct two errors in Section~8 of the above-mentioned paper

concerning the seed parameters proposed for BLS27 elliptic curves

(embedding degree $k = 27$) at the 256-bit and 192-bit security levels.

In both instances, the disclosed seeds produce a composite $r(x)$,

violating the primality condition essential for constructing pairing-friendly curves. Additionally, for the 192-bit seed the

[2026/2011](#) (PDF)

Last updated: 2026-09-28

CauchyFold: Residue-Optimal High-Arity Lattice Folding via Scaled Cauchy Challenges

Xiang Wang

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Cryptographic protocols

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directly creates a mixed term for every pair of inputs. We present CauchyFold, a lattice protocol that folds one accumulator with k fresh quadratic instances without enumerating these pairwise terms.

The companion work [WX26] uses Cauchy folding coefficients to represent the mixed contribution through a vector-valued polynomial of degree below k . We give two algorithms that

[2024/276](#) (PDF)

Last updated: 2026-09-28

Reduce and Prange: Revisiting Prange's ISD for Solving LPN/RSD over Large Fields

Jiseung Kim and Changmin Lee

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Attacks and cryptanalysis

Syndrome decoding (SD), together with its regular-noise variant (RSD), is fundamental to many cryptographic primitives; in the generator formulation used in cryptography, it is closely related to learning parity with noise (LPN).

While recent proposals extend these problems to larger fields, the concrete security of SD over large fields remains comparatively less understood. This gap leaves room for more effective attacks against SD-based primitives over large fields.

In this paper, we present an improved algorithm for solving

[2025/1325](#) (PDF)

Last updated: 2026-09-28

Revisiting the IPA-sumcheck connection

Liam Eagen and Ariel Gabizon

[Hide abstract](#) -

Cryptographic protocols

verifiers, and transparent setup.

Bootle, Chiesa and Sotiraki [BCS21] observed that an IPA can be viewed as a sumcheck protocol [LFKN92] where the summed polynomial is allowed to have coefficients in a group rather than a field. We leverage this viewpoint to

[2026/2247 \(PDF\)](#)

Last updated: 2026-09-28

Geometric Forgeries: Structural Cryptanalysis of MAYO

Massimo Ostuzzi

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Attacks and cryptanalysis

MAYO is a post-quantum signature scheme based on UOV that uses the whipping transformation to achieve more compact public keys and retain efficient signing. We present the first structural forgery attack that exploits the geometry of the resulting multivariate quadratic (MQ) maps. Our starting point is the observation that whipping amplifies polar cancellations in the underlying MQ map, producing subspaces with low-dimensional quadratic and polar images. We combine this geometric insight with the Pseudo-Oil Attack (ASIACRYPT 2026) and a descent to the binary field into a structural forgery algorithm.

[2026/2246 \(PDF\)](#)

Last updated: 2026-09-28

Binding Humans to Keys: A Secure Decentralized Proof of Personhood Protocol

Bilel Zaghdoudi, Gewu BU, Frederic Hayek, Pascal Lafourcade, and Maria Potop-Butucaru

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a digital identity to a unique living human without a trusted central authority, is a foundational problem in permissionless networks. We present a novel protocol, Secure-GYID (S-GYID), resilient to Identity Theft Attack (i.e., adversarial authorities substitute a victim's biometric fingerprint under a malicious public key) and Ghost Key Attack (i.e., colluding authorities fabricate a registration for a non-existent user). Our protocol

[2026/2245 \(PDF\)](#)

Last updated: 2026-09-28

An improved near-capacity lower bound for Reed-Solomon lists on multiplicative subgroups

Al Kindi

[Hide abstract -](#)

Cryptographic protocols

We give an elementary construction of large Reed-Solomon lists on multiplicative subgroups. For every fixed rational rate $\rho \in (0, 1)$, suitable sequences of lengths give lists of size at least $2^{(2-o(1))H(\rho)/\eta}$ at agreement fraction $\rho + \eta$ as $\eta \rightarrow 0$, where H is binary entropy. This doubles the leading entropy exponent of the antipodal lower bound stated by Arnon, Boneh, and Fenzi, building on the construction of Krachun,

[2026/2227 \(PDF\)](#)

Last updated: 2026-09-28

A Fiat-Shamir Transformation for Private-Coin Protocols

Shany Ben-David, Eylon Yogev, and Agni Datta

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Cryptographic protocols

The Fiat--Shamir transformation is a fundamental paradigm in cryptography for compiling interactive public-coin protocols into non-interactive arguments. While this

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verifier's messages depend on secret randomness, remains largely unexplored. To date, there are no candidate transformations for private-coin protocols, neither in the standard model nor in idealized settings.

In this work, we present a generic transformation that compiles any 3-message private-coin interactive proof into a non-interactive argument in the standard model, while preserving zero-knowledge. Our construction relies on sub-

[2026/2244](#) (PDF)

Last updated: 2026-09-28

An Operator-Norm Approach to Security with Quantum Advice

Minki Hhan, Sunghyuk Jo, and Qipeng Liu

[Hide abstract](#) -

Foundations

Non-uniform security allows an adversary to receive bounded advice about an oracle before attempting a fresh challenge. This captures the most realistic attacks and has already been studied extensively in prior work. In this work, we introduce an operator-norm approach for non-uniform security in the quantum random oracle and random permutation models. This new approach yields a unified reduction for both search success probability and distinguishing advantage. Previously, the reduction only worked with success probability even in

[2026/1897](#) (PDF)

Last updated: 2026-09-28

What to Guess in Key-Recovery Attacks?

[Hide abstract](#) -

Secret-key cryptography

Determining the precise parts of the key that need to be guessed in a key-recovery attack is fundamental for judging its cost: if the same attack can be executed by guessing less key material, then the cipher's resistance against this attack is overestimated. Although a multitude of prior works provide upper bounds on the key material required, and although these bounds might be tight in some special cases, a precise evaluation of the required key material and the tightness of

[2026/629](#) (PDF)

Last updated: 2026-09-28

Towards Formal Security Proofs of MQOM

Haruhisa Kosuge and Keita Xagawa

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Public-key cryptography

Recent MPC-in-the-Head (MPCitH) signatures increasingly rely on aggressive GGM-tree optimizations to reduce signature size and cost, culminating in secret-key-root correlated GGM trees as used in SBC (Huth and Joux, CRYPTO 2024), MQOM (NIST PQC Standardization for Additional Signature Round-2, 2024), and rBN++ (Kim, Lee, and Son, EUROCRYPT 2025). While this technique yields substantial compression, it introduces a dependency loop in the proof (Kosuge and Xagawa, ePrint 2025/1999).

We analyze MQOM and resolve this circularity by providing EUF-CMA security proofs for a slightly modified variant. The modifications consist of incorporating a salt into the input of a hash function and modifying the evaluation domain to

TPOKÉ: Threshold Public-Key Encryption from POKÉ via Isogeny Sharing

Toshiki Takatera, Hiroshi Onuki, and Tsuyoshi Takagi

[Hide abstract](#) -

Public-key cryptography

POKÉ, introduced by Basso and Maino, is one of the most efficient isogeny-based public-key encryption schemes. The generic parallel-encryption approach yields a (t, n) -threshold variant by running n independent POKÉ instances and secret-sharing the message, but this multiplies the public-key and ciphertext sizes by roughly n . We present S-TPOKÉ, a POKÉ-based (t, n) -threshold construction for every $1 \leq t \leq n$, and Ch-TPOKÉ, an (n, n) variant that arranges the parties' secret isogenies differently. Both schemes avoid repeating part of the isogeny computation that appears in every independent POKÉ instance. Compared with the trivial construction, they

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Lynx: Symmetric Primitive for Shorter and Faster VOLE-in-the-Head Signatures

Lin Jiao, Hongsen Yang, Hongrui Cui, Yituo He, Yonglin Hao, Xiaojie Guo, Qunxiong Zheng, Jiang Zhang, Yu Yu, and Kang Yang

[Hide abstract](#) -

Secret-key cryptography

VOLE-in-the-Head (VOLEitH) is one of the most promising frameworks to design post-quantum digital signatures based on symmetric primitives. However, all existing symmetric primitives do not capture the specialized characteristics of the

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signatures. In this paper, we propose a VOLEitH-friendly symmetric primitive called Lynx, which is optimal in terms of the number of required VOLE correlations that directly determines the efficiency of VOLEitH-based signature

[2026/2242](#) (PDF)

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Error Propagation-Aware Scale Design for Efficient Homomorphic Encryption-Based LLM Inference

Sieun Seo and Chohong Min

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Applications

Homomorphic encryption (HE) enables privacy-preserving inference by allowing neural networks to operate directly on encrypted data, but its computational cost remains a major obstacle to deploying large language models in practice. In particular, CKKS-based inference consumes ciphertext modulus through homomorphic multiplications and requires costly bootstrapping when the available modulus is exhausted. In this work, we propose an error-propagation-aware scale design for efficient CKKS-based privacy-preserving LLM inference. We characterize the numerical

[2026/477](#) (PDF)

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DAC-PRE: Practical Anonymous Data Access Scheme Control with Proxy Re-encryption for Implantable Medical Devices

Jayaprakash Kar, Xiaoguang Liu, and Fagen Li

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Cryptographic protocols

two primary security components or data access control are typically authorisation and authentication. Data access control is the selective restriction of data access. First, we present an effective data access control mechanism for medical devices that are implanted in this study. Through a

[2026/2241 \(PDF\)](#)

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Quantum Security of XOR of Permutations via Fourier Analysis

Wonseok Choi, Minki Hhan, and Junyoung Jang

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Secret-key cryptography

The XOR of two or more independent random permutations (XoP) is the prototypical pseudorandom function built from permutations achieving the security beyond the birthday bound. The security of the XoP construction is now well established against classical adversaries, however, its security against quantum adversaries that query the construction in superposition has remained widely open.

We prove that the XOR of $r \geq 2$ independent random permutations over $\{0, 1\}^n$ is indistinguishable from a random function by any q -query quantum algorithm with advantage

$$O\left(\min\left\{\frac{q^3}{2^{rn}}, \frac{q^{1.5}}{2^{(r-0.5)n}}, \frac{1}{2^{(r-1.5)n}}\right\}\right)$$

on r . In particular, the XoP construction remains secure throughout the entire query range, far beyond the $2^{n/3}$ quantum birthday bound due to the quantum collision finding attack. To our knowledge, this is the first construction from permutations that achieves the quantum version of the beyond birthday bound security.

[2026/2240 \(PDF\)](#)

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Improved Cryptanalysis of Local PRGs

Shihan Lyu, Mo Li, Chuhan Ma, and Ximing Fu

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Attacks and cryptanalysis

We present two seed-recovery attacks on a family of local PRGs, which map an n -bit random seed to an m -bit pseudorandom string by applying $\backslash\text{XorMaj}$ predicates.

For PRGs with large locality, we extend the Group-and-Solve (GAS, Eurocrypt 26) framework to Guess-Filter-Solve (GFS). GFS first guesses that a set of seed bits are all 1s and then filters for outputs whose inputs to $\backslash\text{Maj}$ contain a sufficiently large subset of the guessed bits. Then high-bias noisy equations can be collected for the correct guess and solved by specific solvers. Compared with GAS, GFS collects substantially more noisy equations with high bias and hence outperforms GAS significantly, especially when the output is short. For example, for the challenge parameter $n = 256$ and $m = 2^{16}$ (STOC 16), the complexity for GFS is $2^{68.78}$ while it is

Zero-Knowledge Proof

Hyeonhak Kim, Won Kim, and Changmin Lee

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Public-key cryptography

Threshold signatures for distributed systems require compact public keys and signatures to reduce communication overhead by avoiding packet fragmentation.

However, with existing post-quantum threshold signatures, either the public key or the signature does not fit within a single unfragmented network packet.

In this work, we present SPRITZ, an isogeny-based post-quantum threshold signature scheme whose public keys and signatures both fit within a single unfragmented network packet at every NIST security level.

To the best of our knowledge, SPRITZ is the first post-quantum threshold signature scheme to do so with an arbitrary number of parties.

While isogeny-based signatures such as SQLsign and PRISM

[2026/2239 \(PDF\)](#)

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One Unverified Coordinate Is Enough: Key Recovery from Faulty Fujisaki–Okamoto Rejection Checks in ML-KEM

Bhabani Sankar Das

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Attacks and cryptanalysis

The Fujisaki-Okamoto (FO) transform makes ML-KEM IND-CCA secure by re-encrypting the decrypted message and returning an implicit-rejection value on any mismatch.

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conditional move (pqc_kyber/cosmian_kyber, RUSTSEC-2026-0290/-0288) and truncated ciphertext comparisons (wolfSSL, CVE-2026-10097 and CVE-2026-6330). We ask how much of the check must be broken before the key is recoverable, and

[2025/1495](#) (PDF)

Last updated: 2026-09-27

Pairwise independence of AES-like block ciphers

Tim Beyne, Gregor Leander, and Immo Schütt

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Secret-key cryptography

We prove that $4r + 4$ rounds of an AES variant with independent and uniform random round keys are ε -close to pairwise independent with $\varepsilon = 2^{14} 2^{-39r}$. This result follows

[2026/2238](#) (PDF)

Last updated: 2026-09-27

Threshold Signatures with Identifiable Abort from One-way Functions

Mojtaba Fadavi, David Jao, and Samuel Jaques

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Cryptographic protocols

A (T, N) -threshold signature scheme distributes a signing key among N participants so that any T participants can jointly generate a valid signature, while fewer than T cannot.

In this work, we first introduce a threshold one-time signature scheme with `\emph{identifiable abort}` based on a variant of Lamport one-time signature and Shamir secret sharing, enabling authorized participants to jointly reconstruct signatures from partial signatures. We then extend the scheme to support multiple messages using multi-layer

2025/2328 (PDF)

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Micali's SNARG, Function Vector Commitments, and Fiat-Shamir

Ziyi Guan and Eylon Yogev

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Foundations

Micali's construction of succinct non-interactive arguments (SNARGs) combines a probabilistically checkable proof (PCP), a vector commitment, and the Fiat-Shamir transformation. Its security is established in the random oracle model, but replacing the random oracle with an explicit hash function presents a fundamental challenge: Fiat-Shamir can fail for interactive arguments, and certain choices of the commitment scheme make Micali's construction unsound for every instantiation of the Fiat-Shamir hash.

We show how to instantiate Micali's construction in the standard model assuming LWE and function vector commitments with $\text{function statistical binding}$ for

2026/2236 (PDF)

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A Complete Classification of Whole-Output Linear Structures in FEILIAN-Type Components

Mounir IDRASSI

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Secret-key cryptography

We determine all input differences with constant whole-output XOR derivatives for a four-word ARX topology containing FEILIAN SubColumn. For every word width $w \geq 2$

family and has dimension at most five, with equality attained. We give a necessary-and-sufficient classifier involving at most 32 candidate masks and characterize all additional structures, including an exceptional coset possible only when the first

[2026/2235](#) (PDF)

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Practical Null-Branch Witness Attacks on In-the-Head Signatures

Guoxiao Liu, Zhuoqing Peng, Guowei Liu, Keting Jia, Xiaoyun Wang, Zongyue Wang, and Junfeng Fan

[Hide abstract](#) -

Attacks and cryptanalysis

In-the-head signatures provide a route to post-quantum authentication based on symmetric primitives. Their algebraic instantiations rely on constraint relations that faithfully encode the underlying one-way function (OWF). We give a classical, public-key-only forgery attack on AIM2-based AIMer v2.0; AIMer was selected in Korea's KpqC competition. For every honestly generated public key of every v2.0 parameter set, the attack forges signatures on arbitrary messages with probability one, without signing queries.

We introduce a \emph{null-branch witness attack} exploiting zero factors that leave intermediate values unconstrained. We complete these local assignments into witnesses satisfying the full relation, including public-output binding, using only public data and without solving an OWF inversion problem.

[2026/2234](#) (PDF)

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Augustin P. SARR

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Uncategorized

In~\cite{sarr25}, Sarr introduced an embarrassingly parallel impersonation attack against ECMQV and ECHMQV-C whose core step is an adding-based pseudo-random walk over a cyclic group G , of order n , searching for a

\emph{decomposed i -point}.

In this work, we revisit that construction in the richer multi-party

setting where two sets of parties,

the initiators $\{\hat{A}_1, \dots, \hat{A}_{z_1}\}$, and the

responders $\{\hat{B}_1, \dots, \hat{B}_{z_2}\}$, interact under ECHMQV-C.

First, we exhibit an embarrassingly parallel attack such that, when m -processors

are available, the attack runs in expected time

$$\frac{\sqrt{n}}{m z_1 z_2} (\backcosta(z_1) + \backcostd(z_1 z_2)),$$

wherein

\backcosta(z_1) is the time for a batch of z_1 elliptic-curve additions

and \backcostd($z_1 z_2$) is the time for a batch of $z_1 z_2$ digest computations.

[2026/2233](#) (PDF)

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Mind the Gap: Proving and Improving RPKI

Shay Cohen, Nicholas Scaglione, Yossi Gilad, Hemi Leibowitz, Bing Wang, and Amir Herzberg

[Hide abstract](#) -

Cryptographic protocols

We present the first rigorous security analysis of the Resource Public Key Infrastructure (RPKI), an IETF standard for protecting inter-domain routing from basic yet effective attacks: prefix and subprefix hijacks. Existing evaluations of RPKI's security focus on empirical studies: adoption measurements, simulations evaluating the security provided by different adoption levels, and identification of implementation and specification flaws.

[2026/2231](#) (PDF)

Last updated: 2026-09-27

Truncated Differential Preimage Attacks via Differential-Linear Correlations

Chunning Zhou, Kai Hu, Zhongfeng Niu, Thomas Peyrin, and Hongyi Zhang

[Hide abstract](#) -

Attacks and cryptanalysis

We derive joint truncated differential (TD) probabilities from multiple differential-linear (DL) approximations and use them in a framework for preimage filtering. For a fixed input difference, the inverse Walsh transform recovers the probabilities of affine output cosets from DL correlations over a complete mask subspace. In our applications, these correlations are estimated using the round-based DL method.

assuming independence among the DL events. The framework handles multiple targets, sharing base evaluations

[2026/1326 \(PDF\)](#)

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LaMS: A p-adic Layered Modulus Switching for Provable Dual Attacks on LWE

Rui-Jie Wang, Zhong-Xiao Wang, Qun-Xiong Zheng, and Xuan Zhao

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Attacks and cryptanalysis

The Learning with Errors (LWE) problem is a central foundation for post-quantum schemes such as Kyber and Dilithium. Dual attacks are among the main tools for assessing the concrete hardness of LWE instances. At EUROCRYPT 2024, Pouly and Shen introduced the first provable dual attack against LWE. Subsequently, at ASIACRYPT 2025, Qu and Xu incorporated modulus switching into this framework by recovering the guessed secret mod several small primes and recombining the resulting residues via the Chinese Remainder Theorem (CRT). Although this CRT-based strategy substantially reduces the search space, it reconstructs the full guessed secret through several distinct primes whose product must exceed q . Consequently, the total guessing cost is dominated by the largest CRT prime p_k . This

[2026/2230 \(PDF\)](#)

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Indistinguishability of Sum of Permutations: A Fourier Analytic Route to Classical and Quantum Security

Ritam Bhaumik, Chun Guo, Xiaoning Guo, and Ashwin Jha

of independent random permutations and related transformations from permutations to functions. Let G be a finite abelian group of order N , and let $\pi_+^k(x) = \pi_1(x) + \cdots + \pi_k(x)$ for $k \geq 2$ independent uniform random permutations of G . We give a unified Fourier analytic treatment in which the construction is represented by its probability density and a distinguisher by its acceptance function, with the classical and quantum query models imposing different restrictions on the Fourier support of the latter.

Classically, we obtain the bound $O_k(q/N^{k-1/2})$ for every



[2026/2229](#) (PDF)

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Local Rewriting under Assumed Erasure

Napassorn Litchiowong

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Foundations

We study local rewriting after an assumed erasure step in an ideal oblivious-transfer protocol. Bob transforms his retained record while Alice's actual record remains fixed. Exact rewriting between shared-source and independent-source

